

SCHOOL OF ENGINEERING

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

BTECH DEGREE II SEMESTER SECOND INTERNAL EXAMINATION, JULY 2024

SUBJECT: 23-200-0205A BASIC ELECTRONICS ENGINEERING (2023 SCHEME)

Duration: 2 HOURS

Maximum Marks: 30

ME A/ SFE

Course Outcomes:

On successful completion of the course, the students will be able to:

- CO 1: Illustrate the concept of diode in rectifiers, filter circuits and wave shaping.
- CO 2: Interpret the functioning of oscillators and operational amplifiers.
- CO 3: Explain the principles of embedded systems and sensors.
- CO 4: Summarize the functioning of a communication system and different modulation technologies

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply,
L4 – Analyze, L5 – Evaluate, L6 – Create

PART - A

(Answer all questions)

(5 x 2 = 10)

Q. No.	Question	Marks	BL	CO	PO
I. a)	State the conditions for oscillation	2	L1	2	1
b)	Explain the need for multiplexing in communication systems, providing an example to illustrate its importance.	2	L2	4	1
c)	Describe the working principle of a transducer and provide examples of an input transducer and an output transducer.	2	L1	3	1
d)	Sketch a circuit diagram of a basic crystal oscillator and explain how the crystal itself determines the oscillation frequency.	2	L2	2	1
e)	In the circuit of a Wien bridge oscillator, if $C_1 = C_2 = 10 \text{ nF}$, determine the output frequencies produced by this arrangement (i) when $R_1 = R_2 = 1 \text{ k}\Omega$ and (ii) when $R_1 = R_2 = 50 \Omega$	2	L3	2	1

PART - B

(2 x 10 = 20)

Q. No.	Question	Marks	BL	CO	PO
II a)	Explain briefly how the astable multivibrator operates. Illustrate your answer using a waveform sketch.	4	L2	2	1
b)	An astable multivibrator circuit is required to produce an asymmetrical rectangular output which has a period of 4ms and is to be high for 1ms and low for 3ms. If the timing capacitors are both to be 100nF, determine the values of the two timing resistors required.	2	L3	2	1
b)	Using circuit diagrams and waveforms, design an operational amplifier circuit to achieve the following functions: a) Obtain the sum of two waveforms. b) Generate a spike for a pulse input.	4	L2	2	1

OR

III	a) Compare and contrast RISC (Reduced Instruction Set Computing) and CISC (Complex Instruction Set Computing) architectures.	5	L1	3	1
	b) Classify embedded systems based on the following criteria: generation, complexity and performance, deterministic behavior, and triggering mechanisms.	5	L1	3	1
IV	a) What is an embedded system? List six points to differentiate between a general-purpose computing system and an embedded system	6	L1	3	1
	b) Differentiate between a microprocessor and a microcontroller	4	L1	3	1

OR

V	a) Describe a basic communication system with a clear block diagram. Identify the major components of the system and explain each one using block diagrams.	10	L1	4	1
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L1=68% L2= 24% L3=8%

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